EDIC & Emergency Critical Care — Clinical Master Reference

Integrated High-Yield Clinical Synthesis: Resuscitation Priorities, Pitfalls, and Standards of Care

This master clinical reference synthesizes the core principles from the study manual from an Emergency & Critical Care consultant's perspective, covering high-stakes resuscitation interventions, clinical traps, and guideline-mandated standards of care.

1. Primary Survey & Life-Saving Interventions

A. Airway:

- **Intubation Decision-Making:** Never base intubation solely on an isolated metric (e.g., GCS alone) or the absence of a gag reflex. It requires a holistic assessment of the patient's capacity to maintain patency, oxygenate, ventilate, and clear secretions.
- **Anatomical Positioning:** In neonates and infants, place a shoulder roll to compensate for the prominent occiput. In obese patients, use the ramped position (aligning the external auditory meatus horizontally with the sternal notch) to optimize functional residual capacity and line-of-sight.
- **Cannot Intubate, Cannot Oxygenate (CICO):** Immediate progression to front-of-neck access via emergency surgical cricothyroidotomy (vertical midline skin incision followed by a horizontal cricothyroid membrane incision). Avoid repeated failed laryngoscopic attempts. In small infants/children, use needle cricothyroidotomy with jet insufflation due to laryngeal cartilaginous fragility.
- **Endotracheal Tube Confirmation:** Continuous quantitative waveform capnography is mandatory; a single point colorimetric check or auscultation alone is insufficient.

B. Breathing & Mechanical Ventilation:

- **Acute Post-Intubation Collapse:** Immediately disconnect the patient from the mechanical ventilator and transition to manual bag-valve-mask ventilation to evaluate chest compliance and differentiate circuit failure from tension pneumothorax or mainstem bronchus intubation.
- **Lung-Protective Ventilation:** Target a low tidal volume of 6 mL/kg predicted body weight (PBW) with a low driving pressure (P_{plat} − PEEP < 14–15 cmH_2O). Accept permissive hypercapnia as long as arterial pH remains > 7.20.
- **Dynamic Hyperinflation (Auto-PEEP):** Persistent expiratory flow at the start of the next inspiratory cycle indicates air trapping. Prolong expiratory time (I:E ratio of 1:4 to 1:5) and reduce the respiratory rate to prevent severe obstructive hemodynamic collapse.

C. Circulation & Shock Resuscitation:

- **High-Quality CPR:** Rate of 100–120 compressions/min, depth of 5–6 cm, full chest recoil, minimal interruptions, and 30:2 ratio prior to advanced airway placement. A sustained abrupt jump in **$ETCO_2$** is an early surrogate of Return of Spontaneous Circulation (ROSC).
- **Special Cardiac Arrest Circumstances:**
  - *Pregnancy > 20 Weeks:* Continuous manual left uterine displacement (LUD) to relieve aortocaval compression. Prepare for resuscitative hysterotomy (perimortem cesarean delivery) if no ROSC occurs within 4 minutes, aiming for delivery by minute 5.
  - *Severe Accidental Hypothermia:* Do not pronounce death prior to rewarming ("not dead until warm and dead"). Withhold IV resuscitation drugs and limit defibrillation attempts to 3 shocks below 30°C until active core rewarming is underway.
- **Distributive / Septic Shock:** Norepinephrine is the first-line vasopressor targeting a mean arterial pressure (MAP) ≥ 65 mmHg. Avoid indiscriminate, excessive crystalloid administration once fluid non-responsiveness is established.
- **Massive Hemorrhage & Damage Control Resuscitation:** Activate Massive Transfusion Protocols (MTP) with a balanced ratio (1:1:1 PRBCs, FFP, platelets). Administer Tranexamic Acid (TXA 1 g IV over 10 min, then 1 g over 8 hours) within 3 hours of trauma.

2. Common Pitfalls to Avoid

- **Airway & Procedural Sedation:**
  - Relying solely on pulse oximetry (**SpO_2**) during procedural sedation; supplemental oxygen delays detection of hypoventilation, whereas capnography detects apnea instantaneously.
  - Failing to recognize progression from moderate to deep sedation when repetitive painful stimulation or manual airway positioning is required.
- **Hemodynamics & Fluid Assessment:**
  - Using static Central Venous Pressure (CVP) to assess volume responsiveness; rely on dynamic maneuvers such as the Passive Leg Raise (PLR) test or end-expiratory occlusion.
  - Excluding occult shock based on a "normal" blood pressure; Shock Index (**SI = HR / SBP > 0.7–0.9**) identifies uncompensated shock prior to frank hypotension.
  - Assuming the absence of pulmonary crackles safely permits volume loading in acute decompensated heart failure.
- **Cardiovascular Emergencies:**
  - Administering AV-nodal blocking agents (adenosine, beta-blockers, non-DHP calcium channel blockers, digoxin, IV amiodarone) to a patient with pre-excited atrial fibrillation (WPW with AF). This shunts impulses down the accessory pathway, causing ventricular fibrillation.
  - Initiating vasodilators prior to achieving heart rate control with beta-blockers in acute aortic syndromes, triggering reflex tachycardia that escalates aortic wall shear stress.
  - Administering nitrates in acute inferior STEMI with right ventricular extension (**ST elevation in V_4R**), which causes profound hypotension due to extreme preload sensitivity.
  - Ruling out acute aortic dissection based on a normal chest radiograph or negative D-dimer in a high-pretest probability clinical scenario.
- **Anaphylaxis:**
  - Administering rapid IV push boluses of concentrated epinephrine (0.5–1 mg) to a perfused patient with a pulse; the first-line route is strictly intramuscular (IM) 1 mg/mL into the anterolateral thigh. IV infusions are reserved for refractory distributive shock.
  - Delaying IM epinephrine to administer antihistamines or corticosteroids, which treat cutaneous signs but do not prevent airway edema or cardiovascular collapse.

3. Standard of Care Guidelines & Clinical Protocols

- **Acute Coronary Syndromes (ACS):** Obtain a 12-lead ECG within 10 minutes of first medical contact. Do not delay primary PCI waiting for troponin results when acute coronary occlusion patterns are present (STEMI, Wellens syndrome, de Winter T-waves, or modified Sgarbossa criteria in LBBB). Fibrinolysis is contraindicated in unstable angina and NSTEMI.
- **Acute Dysrhythmias:** Any unstable tachycardia with hemodynamic compromise, altered mentation, or ischemic chest pain mandates immediate synchronized electrical cardioversion. For symptomatic bradycardia: Atropine 1 mg IV every 3–5 minutes up to 3 mg; transition rapidly to transcutaneous pacing or chronotropic infusions (epinephrine/dopamine) if ineffective or infranodal.
- **Trauma Resuscitation:** Administer TXA within 3 hours. Implement permissive hypotension (target SBP 70–90 mmHg) prior to surgical hemostasis in penetrating or blunt torso hemorrhage without traumatic brain injury. Actively monitor and replete ionized calcium during massive transfusion to prevent citrate toxicity.
- **Hypertensive Emergencies:** Defined by acute or progressing target-organ injury, not arbitrary BP cutoffs alone. In pregnancy, acute severe hypertension (SBP ≥ 160 or DBP ≥ 110 mmHg sustained over 15 minutes) requires IV labetalol, hydralazine, or oral immediate-release nifedipine within 30–60 minutes, with magnesium sulfate for seizure prophylaxis.

4. Neurocritical Care

- **Intracranial Pressure (ICP) Crisis:** Elevate the head of the bed 30 degrees with midline neck positioning to optimize jugular venous return. Administer immediate hyperosmolar therapy: Hypertonic Saline 3% (250–500 mL) or 23.4% (30 mL), or Mannitol (0.5–1 g/kg). Use brief hyperventilation (**$PaCO_2$ 30–35 mmHg**) strictly as a bridge to definitive decompression; prolonged hyperventilation induces severe cerebral ischemia via vasoconstriction.
- **Status Epilepticus Pathway:** First-line (0–5 min): IV Lorazepam 4 mg or IM Midazolam 10 mg. Second-line (5–20 min): IV Levetiracetam 60 mg/kg (max 4500 mg) or Valproate 40 mg/kg. Third-line (refractory, > 20 min): General anesthesia, endotracheal intubation, and continuous infusions of propofol, midazolam, or ketamine with continuous EEG monitoring.
- **Acute Ischemic Stroke:** IV thrombolysis window extends to 4.5 hours; endovascular thrombectomy window extends up to 24 hours based on perfusion imaging mismatch. Lower blood pressure only if > 185/110 mmHg prior to thrombolysis, or > 220/120 mmHg in non-thrombolysis candidates.
- **Pitfalls:** Rapid peripheral phenytoin infusions cause purple glove syndrome and cardiac dysrhythmias (use fosphenytoin). Motor seizure cessation in a comatose patient does not rule out non-convulsive status epilepticus. Aggressive BP lowering in acute stroke compromises penumbral perfusion.

5. Toxicology & Toxidromes

- **Anticholinergic vs. Sympathomimetic Toxidromes:** Both present with tachycardia, mydriasis, and delirium. The crucial differentiator is diaphoresis: **dry skin, dry axillae, and urinary retention** define anticholinergic toxicity, whereas **profuse diaphoresis** characterizes sympathomimetic excess.
- **Specific Antidote Pathways:**
  - *Acetaminophen (Paracetamol):* N-acetylcysteine (NAC) guided by the Rumack-Matthew nomogram; in acute liver failure, continue NAC until coagulopathy and transaminases normalize.
  - *Tricyclic Antidepressants (TCA):* Sodium bicarbonate boluses (1–2 mEq/kg IV) for QRS duration > 100 ms, dysrhythmias, or terminal R wave in aVR ≥ 3 mm.
  - *Calcium Channel & Beta-Blockers:* High-Dose Insulin Euglycemia Therapy (HIET: 1 unit/kg bolus, then 1 unit/kg/hr infusion) accompanied by IV calcium, glucagon, and 20% lipid emulsion therapy for refractory cardiovascular collapse.
  - *Cyanide Poisoning:* Hydroxocobalamin is the first-line antidote; it does not compromise oxygen-carrying capacity.
- **Pitfalls:** Routine flumazenil administration in undifferentiated coma precipitates intractable, life-threatening withdrawal seizures in chronic benzodiazepine users or co-ingested proconvulsants. Gastric lavage is restricted strictly to lethal ingestions presenting within 60 minutes.

6. Electrolytes & Endocrine Emergencies

- **Severe Hyperkalemia (K^+ > 6.5 mEq/L or ECG changes):** Membrane stabilization first: IV Calcium Gluconate 10% (10–20 mL) or Calcium Chloride 10% (via central/wide line in arrest/shock). Intracellular shifting: 10 units regular insulin with 50 mL 50% dextrose, plus high-dose nebulized albuterol (10–20 mg). Elimination: loop diuretics, potassium binders (patiromer, Lokelma), or emergency hemodialysis.
- **Diabetic Ketoacidosis (DKA):** Resuscitate with isotonic/balanced fluids first. **Never initiate insulin if potassium is < 3.3 mEq/L** to prevent fatal dysrhythmias. Add 5% dextrose when serum glucose falls to 200–250 mg/dL while continuing insulin infusion to clear the anion gap.
- **Severe Symptomatic Hyponatremia:** Administer 100–150 mL of 3% hypertonic saline over 10–20 minutes (repeat up to twice) to elevate serum sodium by 4–6 mEq/L and abort cerebral herniation. Avoid chronic correction rates > 8–10 mEq/L in 24 hours to prevent osmotic demyelination syndrome.
- **Pitfalls:** Routine sodium bicarbonate in DKA worsens intracellular acidosis, drives hypokalemia, and increases cerebral edema risk. Ceasing insulin infusion based solely on euglycemia before anion gap closure aborts definitive ketoacidosis resolution.

7. Severe Sepsis & Septic Shock

- Administer empiric broad-spectrum intravenous antimicrobials within the **first hour** of recognizing septic shock (the "Golden Hour").
- Obtain blood cultures prior to antibiotic initiation without delaying therapy beyond 45 minutes.
- Initiate low-dose systemic corticosteroids (Hydrocortisone 200 mg/day) strictly for vasopressor-refractory septic shock requiring escalating doses of norepinephrine and epinephrine.
- Add fixed-dose Vasopressin (0.03 units/min) as a second-line agent to reduce excessive catecholamine dosing rather than driving a single agent to extreme levels.

8. Advanced Trauma & Hemorrhagic Shock

- **Immediately Lethal Thoracic Injuries (The Lethal Six):**
  - *Tension Pneumothorax:* Clinical diagnosis requiring immediate decompression (14G needle in 2nd ICS MCL or 4th/5th ICS AAL, followed by tube/finger thoracostomy). Never delay for radiography.
  - *Cardiac Tamponade:* Beck's triad (hypotension, JVD, muffled heart tones); rapid ultrasound confirmation (eFAST) followed by pericardiocentesis or thoracotomy.
  - *Resuscitative ED Thoracotomy:* Indicated for witnessed cardiac arrest following penetrating chest trauma with signs of life within 10–15 minutes. Goals: pericardiotomy, cardiorrhaphy, open cardiac massage, cross-clamping the descending thoracic aorta.
- **Unstable Pelvic Fractures:** Immediately apply a pelvic binder centered over the **greater trochanters** (not the iliac crests). Never repeatedly assess or "spring" pelvic stability; repeat movement disrupts fragile retroperitoneal clot formation.
- **Severe Traumatic Brain Injury (TBI):** Guard against the secondary insults: hypoxemia (**SpO_2 < 90%**) and hypotension (**SBP < 100–110 mmHg**). A single episode of hypotension significantly increases mortality.
- **Pitfalls:** Resuscitating hemorrhagic shock with normal saline exacerbates hyperchloremic metabolic acidosis, dilutes coagulopathy factors, and accelerates the lethal triad (hypothermia, acidosis, coagulopathy).

9. Acute Respiratory Failure & Mechanical Ventilation

- **ARDS Protective Strategy:** Low tidal volumes (4–6 mL/kg PBW), plateau pressure **$P_{plat} \leq 30\ cmH_2O$**, driving pressure **P_{plat} – PEEP < 14–15 cmH_2O**, and early prone positioning (≥ 16 hours/day) for severe hypoxemia (**$PaO_2/FiO_2 < 150$**).
- **Near-Fatal Asthma:** Warning signs of imminent collapse: silent chest, exhaustion, altered sensorium, and pseudonormalization or elevation of **$PaCO_2$**. Resuscitate with continuous nebulized beta-agonists/anticholinergics, IV steroids, magnesium sulfate 2 g, and IM epinephrine.
- **Mechanical Pitfalls:** Rushing to intubate severe asthmatics carries extreme arrest risks from high intrathoracic pressure and auto-PEEP. If cardiovascular collapse occurs post-intubation, disconnect the ventilator and manually compress the chest to allow passive exhalation.

10. Acute Abdominal & GI Emergencies

- **Massive Upper GI Bleeding:** Early endotracheal intubation for airway protection prior to massive hematemesis. Use restrictive blood transfusion targets (Hb 7–8 g/dL) to prevent portal pressure spikes that worsen variceal rebleeding. In suspected varices: Octreotide (50 mcg bolus, then 50 mcg/hr), prophylactic Ceftriaxone 1 g IV, and erythromycin to improve endoscopic view.
- **Acute Mesenteric Ischemia:** Severe abdominal pain out of proportion to physical exam findings with elevated lactate. Urgent CT mesenteric angiography is the gold-standard diagnostic tool.
- **Pitfalls:** Withholding opioids from acute abdomen patients is obsolete; analgesia facilitates a more accurate examination. A normal serum lactate does not exclude early intestinal ischemia.

11. Environmental Emergencies

- **Heat Stroke:** Core hyperthermia (> 40°C) with central nervous system dysfunction. Gold standard: immediate, whole-body cold-water immersion. Antipyretics are useless and harmful because the hypothalamic set point is unchanged.
- **Severe Accidental Hypothermia:** Gentle patient handling is mandatory; the myocardium is irritable and prone to ventricular fibrillation. Rewarm internally: warmed IV fluids (40–42°C), humidified warm oxygen, and ECMO/ECLS in severe cardiovascular collapse.

12. Pediatric & Neonatal Emergencies

- **Pediatric Assessment Triangle (PAT):** Rapid observational triad: Appearance, Work of Breathing, and Circulation to Skin.
- **Neonatal Resuscitation:** Positive Pressure Ventilation (PPV) at 40–60 breaths/min is the most effective initial intervention for heart rate < 100 bpm or apnea. Initiate chest compressions strictly if heart rate remains < 60 bpm after 30 seconds of effective ventilation (3:1 ratio).
- **Severe Croup vs. Epiglottitis:** Severe croup: Dexamethasone (0.6 mg/kg) plus nebulized racemic/regular epinephrine (0.5 mL/kg of 1:1000, max 5 mL). Epiglottitis: avoid throat examination with tongue depressors; transfer calmly to the operating room for definitive airway control.
- **Fluid Boluses:** Standard resuscitation: 20 mL/kg balanced crystalloid over 10–20 minutes; restrict to 10 mL/kg in suspected cardiogenic shock, DKA, or severe acute malnutrition.
- **Pitfalls:** Bradycardia in children is hypoxemic until proven otherwise; treat ventilation and oxygenation before cardiac drugs. Ceftriaxone in jaundiced neonates displaces bilirubin, causing kernicterus and calcium precipitates.

13. Obstetric & Gynecologic Emergencies

- **Postpartum Hemorrhage (PPH):** The "4Ts" framework (Tone 70%, Trauma 20%, Tissue 10%, Thrombin 1%). For uterine atony: bimanual uterine compression, Oxytocin first-line, misoprostol, methylergonovine (contraindicated in hypertension), carboprost (contraindicated in asthma), and TXA 1 g within 3 hours.
- **Preeclampsia with Severe Features & Eclampsia:** Magnesium sulfate (4–6 g IV loading over 15–20 min, then 1–2 g/hr) for seizure prophylaxis. Control blood pressure < 160/110 mmHg with IV labetalol, hydralazine, or oral nifedipine.
- **Ruptured Ectopic Pregnancy:** Any female of childbearing age with abdominal pain or syncope has an ectopic pregnancy until proven otherwise. Immediate bedside ultrasound to check for free fluid; call surgical gynecology without waiting for quantitative lab results.
- **Pitfalls:** Supine positioning in pregnancy > 20 weeks causes inferior vena cava compression, reducing cardiac output by 30–40%. Methylergonovine in hypertensive patients precipitates intracranial hemorrhage.

14. Hematology & Oncology Emergencies

- **Febrile Neutropenia:** Temperature ≥ 38.3°C with ANC < 500/μL. Administer antipseudomonal beta-lactams (piperacillin-tazobactam, cefepime, or meropenem) within 60 minutes without waiting for diagnostic workups.
- **Tumor Lysis Syndrome (TLS):** Hyperkalemia, hyperphosphatemia, hyperuricemia, and hypocalcemia. Resuscitate with aggressive IV hydration and rasburicase.
- **Malignant Spinal Cord Compression:** New focal back pain with motor deficits or sphincter dysfunction in cancer patients. Immediate high-dose dexamethasone (10–16 mg IV), urgent whole-spine MRI, and emergency neurosurgery/radiation oncology consultation.
- **Pitfalls:** Correcting asymptomatic hypocalcemia in TLS triggers calcium-phosphate precipitation, renal shutdown, and cardiac arrest. Performing digital rectal examinations in neutropenia risks mucosal tear and fulminant bacteremia.

15. Procedures, Ethics & Systems

- **Finger Thoracostomy:** Preferred over needle decompression in ventilated trauma patients to achieve immediate, reliable decompression of tension hemopneumothorax.
- **Rapid Sequence Intubation (RSI):** Optimize preoxygenation to **SpO_2 > 95%** with flush-rate oxygen; maintain apneic oxygenation via nasal cannula. Succinylcholine is contraindicated in pre-existing hyperkalemia, denervation, and burns/crush injuries after 72 hours; rocuronium is the primary alternative with sugammadex reversal.
- **Pitfalls:** Prolonging futile CPR in unwitnessed asystolic/PEA out-of-hospital cardiac arrest without reversible causes. Failing to document the loss of capacity and imminent threat to life when executing emergency implied consent.

16. Musculoskeletal & Rheumatologic Emergencies

- **Acute Compartment Syndrome:** Severe pain out of proportion to the injury, markedly exacerbated by passive stretch of muscles, is the earliest and most sensitive sign. When delta pressure (**$\Delta P = DBP - P_{comp} \leq 30$ mmHg**), emergent decompressive fasciotomy is indicated.
- **Rhabdomyolysis:** Serum CK > 5 times upper limit, tea-colored urine positive on dipstick for blood but negative for RBCs under microscopy. Aggressive fluid resuscitation targeting urine output 2–3 mL/kg/hr prevents myoglobinuric acute tubular necrosis.
- **Septic Arthritis:** Arthrocentesis is mandatory for any acute, hot, swollen, non-weight-bearing joint. Synovial WBC > 50,000/μL with > 75% neutrophils mandates immediate IV antibiotics and orthopedic surgical joint drainage.
- **Pitfalls:** Applying a circumferential plaster cast to a fresh fracture causes acute compartment syndrome. Deferring arthrocentesis because a patient has a history of gout misses co-existent crystal and septic arthritis.

17. Dermatologic Emergencies

- **Necrotizing Soft Tissue Infections (NSTI):** Pain out of proportion to skin findings, rapid progression, bullae, skin anesthesia, and crepitus. Surgical debridement is the only life-saving intervention alongside broad-spectrum coverage (Meropenem + Vancomycin + Clindamycin to inhibit toxin synthesis).
- **SJS / Toxic Epidermal Necrolysis (TEN):** Mucosal involvement with epidermal detachment (positive Nikolsky sign). Discontinue offending medications immediately, provide burn-center level fluid/temperature support, and avoid adhesive dressings.
- **Pitfalls:** Excluding necrotizing fasciitis based on the absence of subcutaneous gas on X-ray; gas is late and absent in group A streptococcal infections. Systemic corticosteroids in late TEN increase sepsis mortality.

18. Ophthalmologic Emergencies

- **Acute Angle-Closure Glaucoma:** Injected eye, severe ocular/orbital pain, corneal clouding, fixed mid-dilated pupil, and intraocular pressure > 40–50 mmHg. Lower IOP with topical timolol 0.5%, apraclonidine 1%, pilocarpine 1–2%, alongside systemic acetazolamide or mannitol.
- **Central Retinal Artery Occlusion (CRAO):** Sudden, painless monocular visual loss; fundus shows pale retina with a cherry-red spot at the macula. Treated as an emergent stroke equivalent.
- **Chemical Eye Burns:** Immediate, copious Morgan-lens eye irrigation with normal saline or lactated Ringer's before history taking or visual acuity testing, continuing until conjunctival pH reaches 7.0–7.4. Alkali burns are more devastating due to liquefactive necrosis.
- **Pitfalls:** Discharging patients with topical anesthetic drops causes corneal ulceration and permanent blindness. Measuring IOP with tonometry in suspected open globe injuries causes extrusion of intraocular contents.

19. Disaster Medicine & Mass Casualty Triage (START)

- **START Triage Pathway:**
  - Sort the "Walking Wounded" and categorize as Green / Minor.
  - Respirations: If absent, open the airway; if still absent, categorize as Black (Deceased). If spontaneous breathing resumes or rate is > 30/min, categorize as Red (Immediate).
  - Perfusion: Capillary refill > 2 sec or absent radial pulse: categorize as Red; control external bleeding immediately.
  - Mental Status: Inability to obey simple commands: categorize as Red; ability to obey: Yellow (Delayed).
- **Ethical Shift:** In mass casualties, medical ethics shifts from "maximum care for the individual" to "doing the greatest good for the greatest number."

20. High-Risk Endocrine Crises

- **Thyroid Storm:** Four-step treatment in strict sequence:
  1. Control adrenergic surge: IV Propranolol (1–2 mg slowly).
  2. Inhibit new hormone synthesis: Propylthiouracil (PTU 500–1000 mg) or Methimazole.
  3. Block preformed hormone release: Inorganic iodine (Lugol's) administered **at least 1 hour after** PTU to avoid fueling further synthesis (Jod-Basedow effect).
  4. Adrenal support & peripheral block: IV Hydrocortisone (100 mg every 8 hours).
- **Acute Adrenal Crisis:** Vasopressor-refractory hypotension with hyponatremia and hyperkalemia. Administer IV Hydrocortisone 100 mg immediately; do not delay for ACTH stimulation testing. (If testing is mandatory, use Dexamethasone 4 mg, which does not cross-react with cortisol assays).
- **Myxedema Coma:** Severe hypothermia, bradycardia, and obtundation. Gentle passive rewarming, IV Levothyroxine (300–500 mcg), and mandatory concurrent hydrocortisone to prevent precipitating fatal adrenal crisis.
- **Pitfalls:** Administering iodine before thionamides in thyroid storm; aggressive active surface warming in myxedema coma causes massive vasodilation and circulatory collapse.

21. Post-Cardiac Arrest Care & Targeted Temperature Management (TTM)

- **Targeted Temperature Management:** Strict fever avoidance (**Core Temp < 37.5°C**) or maintaining a target between 32°C and 36°C for 24–72 hours in comatose patients post-ROSC.
- **Hemodynamic Targets:** Maintain SBP ≥ 100 mmHg and MAP ≥ 65–80 mmHg. Avoid hyperoxia (target **SpO_2 92–98%**) and maintain normocapnia (**$PaCO_2$ 35–45 mmHg**). Immediate emergent coronary angiography is mandatory for post-ROSC STEMI.
- **Pitfalls:** Premature neuroprognostication; definitive neurological prognostication should not occur earlier than 72 hours post-ROSC and only after complete clearance of sedatives, paralytics, and hypothermia.

22. Extracorporeal Life Support & Critical Care Transport

- **Extracorporeal CPR (ECPR):** Indicated for selected young patients (< 65 years) with witnessed arrest, high-quality bystander CPR, initial shockable rhythm (VF/VT) or reversible etiology, and failure of conventional CPR within 15–30 minutes (targeting flow time < 60 min).
- **ECMO Modalities:** VA-ECMO provides biventricular and respiratory support in cardiogenic shock; VV-ECMO provides isolated gas exchange in refractory ARDS with preserved cardiac function.
- **Aero-Medical Transport & Boyle's Law:** Gas expansion at altitude requires replacing air in ETT cuffs with sterile saline, placing a nasogastric tube to decompress the stomach, and proactively placing a chest tube for small pneumothoraces prior to flight.

23. Undifferentiated Cases, Cognitive Biases & Guideline Conflicts

- **Undifferentiated Shock & RUSH Protocol:** Systematic evaluation of the Pump (LV contractility, RV strain, pericardial effusion), the Tank (IVC dynamics, eFAST free fluid, lung sliding), and the Pipes (aortic aneurysm dissection, DVT).
- **Delta-Delta Ratio in Complex Metabolic Acidosis (ΔAG / ΔHCO_3^-):**
  - < 0.8: Mixed High Anion Gap (HAGMA) and Normal Anion Gap (NAGMA).
  - 0.8 – 2.0: Pure High Anion Gap Metabolic Acidosis.
  - > 2.0: Concurrent Metabolic Alkalosis (e.g., vomiting) with HAGMA.
- **Cognitive Biases:** Anchoring bias, availability bias, and premature closure are best mitigated by cognitive forced pauses: *"What is the worst-case fatal diagnosis, and has it been safely excluded?"*
- **Guideline Conflicts:**
  - *IM Epinephrine in Anaphylaxis:* American guidelines recommend 0.3 mg (up to 0.5 mg in severe/heavy patients), whereas European/UK guidelines standardize on 0.5 mg for all adults; higher doses achieve therapeutic plasma levels faster in vasodilatory shock.
  - *Sepsis Fluid Resuscitation:* The traditional 30 mL/kg recommendation is challenged by restrictive practices favoring early norepinephrine initiation to protect heart failure and renal failure patients from volume overload.
  - *Recent-Onset Atrial Fibrillation:* A delayed wait-and-see strategy with rate control over 48 hours allows spontaneous cardioversion in > 70% of stable patients without unnecessary electrical shocks.

24. Niche & High-Acuity Procedures

- **Ultrasound-Guided Pericardiocentesis:** The blind subxiphoid approach has been superseded by ultrasound-guided apical or parasternal approaches targeting the largest, most superficial fluid pocket.
- **Scalpel-Finger-Bougie Cricothyroidotomy:** Standardized technique: #10 scalpel for a horizontal cricothyroid incision, index finger to maintain the tract, bougie introduced into the trachea over the finger, and a cuffed 6.0 mm ETT railroaded over the bougie.
- **Balloon Tamponade in Variceal Bleeding (Sengstaken-Blakemore):** Endotracheal intubation is mandatory before tube insertion. Inflate the gastric balloon first (250–300 mL air, secured under traction); inflate the esophageal balloon strictly if bleeding persists, keeping pressure < 30–45 mmHg to avoid esophageal necrosis.

25. Advanced ECG Nuances & STEMI Equivalents

- **de Winter T-Waves:** 1–3 mm upsloping ST depression at the J-point in precordial leads with tall, prominent, symmetric T waves and ST elevation in aVR. Indicates acute proximal LAD occlusion and mandates immediate primary PCI.
- **Wellens' Syndrome:** Deeply inverted symmetric T waves (Type B) or biphasic T waves (Type A) in leads V2–V3 in a pain-free patient with normal cardiac enzymes. Represents critical proximal LAD stenosis; stress testing is strictly contraindicated.
- **Modified Sgarbossa Criteria in LBBB:** Excessively discordant ST elevation where ST elevation is ≥ 1 mm and constitutes > 25% of the preceding S wave depth (**ST/S ratio ≤ −0.25**) accurately identifies acute coronary occlusion.

26. Differential Trap Matrix

Clinical Condition	Pathophysiological Mechanism	Distinguishing Signs	Definitive Treatment Protocol
TRALI (Transfusion-Related Acute Lung Injury)	Non-cardiogenic alveolar infiltrate and neutrophil activation triggered by donor antibodies.	Normal or low BP, no JVD, normal BNP, protein-rich tracheal aspirate.	Supportive lung-protective ventilation, discontinue transfusion immediately, notify blood bank. *Diuretics contraindicated.*
TACO (Transfusion-Associated Circulatory Overload)	Cardiogenic hydrostatic pulmonary edema caused by rapid or excessive volume loading.	Acute hypertension, pronounced JVD, S3 gallop, marked BNP elevation.	Stop transfusion immediately, administer IV loop diuretics (furosemide), and preload-reducing nitrates.
Histaminergic Angioedema	IgE-mediated mast cell and basophil degranulation with histamine release.	Rapid onset, accompanied by pruritus, urticaria, and generalized flushing.	Responds to IM epinephrine, H1/H2 receptor blockers, and corticosteroids.
Bradykinin-Mediated Angioedema	Bradykinin accumulation due to ACE-inhibitors or C1-inhibitor deficiency (HAE).	Slower progression, **no pruritus, no urticaria**, predilection for lips, tongue, and bowel wall.	**Completely refractory to epinephrine, antihistamines, and steroids.** Treat with Icatibant, C1-inhibitor concentrate, or FFP.

> This master clinical reference covers the complete theoretical, algorithmic, and numerical framework of the EDIC study guide in a unified, print-ready English layout.